1 ое задание

I. Неявные функции

Т1. – разобрали на семинаре

§3 №61.2.

$u(x,y)$

$x\cos y + y\cos u + u\cos x = 1 \quad (0,1,0) = (x,y,u)$

$F = x\cos y + y\cos u + u\cos x - 1 = 0.$

$F'_x = \cos y + u(-\sin x)$. Подставляю: $F'_x = \cos 1 - 0\cdot\sin 0 = \cos 1,$

$F'_x \neq 0.$

$F'_y = -\sin y \cdot x + \cos u$ Подставляю: $F'_y = 0 + \cos 0 = 1. \neq 0.$

$F'_u = -\sin u \cdot y + \cos x$ Подставляю $F'_u = -\sin 0 + \cos 0 = 1 \neq 0.$

$u'_x = -\frac{F'_x}{F'_u} = -\frac{\cos 1}{1} = -\cos 1 \qquad u'_y = -\frac{F'_y}{F'_u} = -1.$

Ответ: $\frac{\partial u}{\partial x} = -\cos 1; \quad \frac{\partial u}{\partial y} = -1.$

64.1. du – ? $u(x,y)$

$u^3 - xu + y = 0$

$F(x,y,u) = 0$

а) $(3;-2;2)$

б) $(3;-2;-1).$

$F'_x = -u$ а) $F'_x = -2$ б) $F'_x = 1.$ | $F'_y = 1$ | $F'_u = 3u^2 - x$ а) $F'_u = 9$ б) $F'_u = 0.$

$u'_x = -\frac{F'_x}{F'_u}$ а) $u'_x = \frac{2}{9}$

$u'_y = -\frac{F'_y}{F'_u}$ а) $u'_y = -\frac{1}{9}.$

Ответ: а) $du = \frac{2}{9}dx - \frac{1}{9}dy.$

б) нельзя.

№72. $f(x-az, y-bz) = 0$ $f(u,v)$ – дифф

a, b – const.

$z(x,y)$:

$$a\frac{\partial z}{\partial x} + b\frac{\partial z}{\partial y} = 1.$$

Решение: Пусть: $u = x - az$

$v = y - bz$, тогда $f(u,v) = 0$.

$$\frac{df}{dx} = \frac{df}{du}\cdot\frac{du}{dx} + \frac{df}{dv}\frac{dv}{dx} = 0. \quad \text{Подставляем:}$$

$$\frac{df}{dy} = \frac{df}{du}\frac{du}{dy} + \frac{df}{dv}\cdot\frac{dv}{dy} = 0.$$

$$\frac{df}{dx} = \frac{df}{du}\frac{\partial}{\partial x}(x-az) + \frac{df}{dv}\frac{\partial}{\partial x}(y-bz) = \frac{df}{du}\left(1 - a\frac{\partial z}{\partial x}\right) + \frac{df}{dv}\left(-b\frac{\partial z}{\partial x}\right) = 0.$$

$$\frac{df}{dy} = \frac{df}{du}\frac{\partial}{\partial y}(x-az) + \frac{df}{dv}\frac{\partial}{\partial y}(y-bz) = \frac{df}{du}\left(-a\frac{\partial z}{\partial y}\right) + \frac{df}{dv}\left(1 - b\frac{\partial z}{\partial y}\right) = 0.$$

$\frac{\partial z}{\partial x}$: $f'_u - af'_u\frac{\partial z}{\partial x} - bf'_v\frac{\partial z}{\partial x} = 0$. $\Rightarrow \frac{\partial z}{\partial x} = \frac{f'_u}{a\cdot f'_u + b\cdot f'_v}$.

$\frac{\partial z}{\partial y}$: $-af'_u\cdot\frac{\partial z}{\partial y} + f'_v - bf'_v\cdot\frac{\partial z}{\partial y} = 0 \Rightarrow \frac{\partial z}{\partial y} = \frac{f'_v}{bf'_v + af'_u}$.

$$a\cdot\frac{\partial z}{\partial x} + b\frac{\partial z}{\partial y} = a\cdot\frac{f'_u}{a\cdot f'_u + b\cdot f'_v} + b\frac{f'_v}{bf'_v + a\cdot f'_u} = \frac{af'_u + bf'_v}{af'_u + bf'_v} =$$

$= 1$. Ч.Т.Д.

№75. Найти Я.П. $u(x;y)$, $v(x;y)$ $(x,y,u,v) = (1,2,0,0)$

$F(x;y)$.

$\frac{\partial f}{\partial x} = -\left[\frac{\partial F}{\partial y}\right]^{-1}\left[\frac{\partial f}{\partial x}\right]$

У нас $\partial x \approx (x;y)$

$\partial y \approx (u;v)$

Найдем $\frac{\partial F}{\partial y} = \begin{pmatrix} \frac{\partial F_1}{\partial u} & \frac{\partial F_1}{\partial v} \\ \frac{\partial F_2}{\partial u} & \frac{\partial F_2}{\partial v} \end{pmatrix}$.

$\frac{\partial F}{\partial x} = \begin{pmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} \end{pmatrix}$.

$F_1 = x\cdot e^{u+v} + 2uv - 1 = 0$ $\quad F_2 = ye^{u-v} - \frac{u}{1+v} - 2x = 0$.

$\frac{\partial F_1}{\partial u} = xe^{u+v} + 2v = \Big|_{(1,2,0,0)} = 1$

$\frac{\partial f_1}{\partial v} = x e^{u+v} + 2u \Big|_{(1,2,0,0)} = 1$

$\frac{\partial f_2}{\partial u} = y \quad e^{u-v} - \frac{1}{1+v} \Big|_{1,2,0,0} = +1.$

$\frac{\partial f_2}{\partial v} = -ye^{u-v} + \frac{u}{(1+v)^2} \Big|_{1,2,0,0} = -2$

$\frac{\partial f}{\partial y} = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}$ $\det = -3$

$\left(\frac{\partial f}{\partial y}\right)^{-1} = -\frac{1}{3}\begin{pmatrix} -2 & -1 \\ -1 & 1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}.$

$\frac{\partial f_1}{\partial x} = e^{u+v} \Big|_{1,2,0,0} = 1 \qquad \frac{\partial f_2}{\partial x} = -2.$

$\frac{\partial f_1}{\partial y} = 0 \qquad \frac{\partial f_2}{\partial y} = e^{u-v} = 1.$

$\frac{\partial f}{\partial x} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$

$\begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = \frac{1}{-3}\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 3 & -1 \end{pmatrix} \cdot \frac{1}{3}\begin{pmatrix} -1 \end{pmatrix}.$

§4 №43.6.

$d^2u - ?$ $\quad u(x,y)$ $\quad M_0(x_0 y_0)$

$y - u = e^{xu}; \quad u(1;1) = 0.$

$\Rightarrow \quad (1;1;0) \quad (x,y,0) \qquad F = y - u - e^{xu} = 0.$

Т.к. $x;y$ - независимые.

$d^2u = \frac{\partial^2 u}{\partial x^2}dx^2 + 2\frac{\partial^2 u}{\partial x \partial y}dx dy + \frac{\partial^2 u}{\partial y^2}dy^2.$

$F'_x = -e^{xu} \cdot u \Big|_{(1;1;0)} = 0 \qquad u'_x = -\frac{F'_x}{F'_u} = 0$

$F'_y = 1 \qquad u'_y = -\frac{F'_y}{F'_u} = \frac{1}{2}$

$F'_u = -1 - e^{xu} \cdot x \Big|_{(1;1;0)} = -2.$

$$u''_{xx} = \left(-\frac{e^{xu}\cdot u}{1+e^{xu}\cdot x}\right)'_x = \left(\frac{-e^{x\,u(x;y)}\cdot u(x;y)}{1+e^{x\cdot u(x;y)}\,x}\right)'_x =$$

$$= \frac{\left(u'(x;y)\cdot(-e^{x\cdot u}) + u(x;y) - e^{x\cdot u(x;y)}\left(u(x;y)+x\,u'(x;y)\right)\right)(1+e^{xu}x)+}{(1+xe^{xu})^2}$$

$$+\frac{u\cdot e^{xu}\left(e^{x\cdot u}+x\cdot e^{xu}(u+xu')\right)}{} = 0.$$

$$u''_{yy} = \left(+\frac{1}{1+xe^{xu}}\right)'_y = -1\,\frac{(xe^{xu}\cdot u'_y)}{(1+xe^{xu})^2} = -\frac{\frac{1}{2}\cdot 1\cdot 1}{(1+1)^2} = -\frac{1}{8}$$

$$u''_{xy} = \left(\frac{1}{1+xe^{xu}}\right)'_x = -\frac{e^{xu}+xe^{xu}(u+xu')}{(1+xe^{xu})^2} = -\frac{1+1\cdot 1(0+0)}{4} = -\frac{1}{4}$$

Ombem: $-\frac{1}{2}dxdy - \frac{1}{8}dy^2$.

46.2. $f\left(\frac{x}{u};\frac{y}{u}\right)=0$ $\quad d^2u(x;y)-?$

$u = f(v;w)$.

$du = f'_v dv + f'_w dw$

$\frac{x}{u} = v \quad \frac{y}{u} = w$

$dv = \frac{dx}{u} - \frac{x}{u^2}du \quad dw = \frac{dy}{u} - \frac{y}{u^2}du$

$du = f'_v dv + f'_w dw$

$f'_v\left(\frac{dx}{u} - \frac{x}{u^2}du\right) + f'_w\left(\frac{dy}{u} - \frac{y}{u^2}du\right) = du \quad D = Ax+By$

$$du = \frac{f'_v\frac{dx}{u} + f'_w\frac{dy}{u}}{1+f'_v\frac{x}{u^2}+f'_w\frac{y}{u^2}} = u\,\frac{f'_v dx + f'_w dy}{(u^2+f'_v x + f'_w y)} = u\,\frac{Adx+Bdy}{u^2+D}$$

$u'_x = \frac{u\,A}{u^2+\Theta}$ $\quad u'_y = \frac{B u}{u^2+\Theta}$ $\quad \Theta' = A + x(A'_v v'_x + A'_w w'_x) + y(B'_v v'_x + B'_w w'_x)$

$$u''_x = \frac{(u'_x A + u(A'_v v'_x + A'_w w'_x))(u^2+\Theta) - uA(2uu' + \Theta')}{(u^2+\Theta)^2} =$$

§3 №106.1

Найти Якобиан $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ отображения:

$u = xyz$

$v = xy - xyz$

$w = y - xy.$

$$J = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{pmatrix} = \begin{pmatrix} yz & xz & xy \\ y-yz & x-xz & -xy \\ -y & 1-x & 0 \end{pmatrix} =$$

$\frac{\partial u}{\partial x} = (xyz)' = yz. \quad \frac{\partial u}{\partial y} = xz \quad \frac{\partial u}{\partial z} = xy$

$\frac{\partial v}{\partial x} = y - yz \quad \frac{\partial v}{\partial y} = x - xz \quad \frac{\partial v}{\partial z} = -xy$

$\frac{\partial w}{\partial x} = -y \quad \frac{\partial w}{\partial y} = 1 - x \quad \frac{\partial w}{\partial z} = 0.$

$= yz(xy(1-x)) - xz(-y \cdot xy) + xy((y-yz)(1-x) + y(x-xz)) =$

$= xy^2z - x^2y^2z + x^2y^2z + xy(y - xy - yz + xyz + xy - xyz) =$

$= xy^2z + xy^2 - xy^2z = xy^2.$ Ответ: xy^2.

ТЗ. $f: \mathbb{R}^2 \to \mathbb{R}^2$:

$u = e^x \cos y$

$v = e^x \sin y.$

$$J = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = \begin{pmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{pmatrix}$$

$= e^{2x}\cos^2 y + e^{2x}\sin^2 y = e^{2x}(\ ) = e^{2x} \neq 0.$ $\Rightarrow$ Отличен от 0.

$\begin{pmatrix} u \\ v \end{pmatrix}(x;y) = \begin{pmatrix} u \\ v \end{pmatrix}(x, y+2\pi) = ux + v(y+2\pi).$

не взаимнооднозначно на $\mathbb{R}^2 \setminus (0;0)$.

$\mathbb{R}^2 \setminus (0;0) \longleftrightarrow (-\infty, +\infty) \times [-\pi; \pi]$

Тч. $f: \mathbb{R}^2 \to \mathbb{R}^2$ $\quad x = r\cos\varphi$, $y = r\sin\varphi$; $r > 0$.

$x^2 + y^2 = r^2$

$r = \sqrt{x^2+y^2}$.

$\operatorname{tg}\varphi = \frac{y}{x}$.

$J = \begin{pmatrix} r'_x & r'_y \\ \varphi'_x & \varphi'_y \end{pmatrix}$ $\quad \frac{\partial r}{\partial x} = \frac{1}{2\sqrt{x^2+y^2}} \cdot 2x = \frac{x}{\sqrt{x^2+y^2}} = \frac{x}{r}$.

$\frac{\partial r}{\partial y} = \frac{1 \cdot 2y}{2\sqrt{x^2+y^2}} = \frac{y}{\sqrt{x^2+y^2}} = \frac{y}{r}$

$\operatorname{arctg}\operatorname{tg}\varphi = \operatorname{arctg}\frac{y}{x}$

$\varphi = \operatorname{arctg}\frac{y}{x}$

$\frac{d\varphi}{dx} = \frac{1}{1+\left(\frac{y}{x}\right)^2} \cdot (-1)\frac{y}{x^2} = -\frac{y}{x^2+y^2} = -\frac{y}{r^2}$

$(\operatorname{arctg} x)' = \frac{1}{1+x^2}$

$\frac{d\varphi}{dy} = \frac{1}{1+\left(\frac{y}{x}\right)^2} \cdot \frac{1}{x} = \frac{x}{x^2+y^2} = \frac{x}{r^2}$

$J = \begin{pmatrix} \frac{x}{r} & \frac{y}{r} \\ -\frac{y}{r^2} & \frac{x}{r^2} \end{pmatrix}$.

$y = f(x)$ $\quad x = g(y)$.

$J_f = J^{-1}_g$.

$x = x(r, \varphi)$

$y = y(r, \varphi)$. $\quad \frac{\partial x}{\partial r} = \cos\varphi$; $\frac{\partial x}{\partial \varphi} = -r\sin\varphi$.

$\frac{\partial y}{\partial r} = \sin\varphi$; $\frac{\partial y}{\partial \varphi} = r\cos\varphi$.

$J(r,\varphi) = \begin{pmatrix} \cos\varphi & -r\sin\varphi \\ \sin\varphi & r\cos\varphi \end{pmatrix}$. $\quad J^{-1}_{(r,\varphi)} = \frac{1}{r}\begin{pmatrix} r\cos\varphi & r\sin\varphi \\ -\sin\varphi & \cos\varphi \end{pmatrix} =$

$= \begin{pmatrix} \cos\varphi & \sin\varphi \\ \frac{-\sin\varphi}{r} & \frac{\cos\varphi}{r} \end{pmatrix}$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{\Delta}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

$$\begin{pmatrix} \frac{x}{r} & \frac{y}{r} \\ -\frac{y}{r^2} & \frac{x}{r^2} \end{pmatrix}.$$

$\det J(r,\varphi) = r\cos^2\varphi + r\sin^2\varphi = r.$

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§3 №86. $x\frac{\partial u}{\partial y} - y\frac{\partial u}{\partial x} = 0.$

$F(x,y,u)=0.$

Полярные

$x = r\cos\varphi$
$y = r\sin\varphi.$

$x^2+y^2=r^2$

$\varphi = 1.$

$\operatorname{tg}\varphi = \frac{y}{x}.$

$u = u(r(x,y), \varphi(x,y))$

$$\frac{\partial u}{\partial x} = \frac{\partial u(r(x,y),\varphi(x,y))}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial \varphi}\frac{\partial \varphi}{\partial x} = \frac{\partial u}{\partial r}\frac{x}{r} - \frac{\partial u}{\partial \varphi}\frac{y}{r^2}.$$

$\frac{\partial r}{\partial x} = \frac{x}{r} \quad \frac{\partial \varphi}{\partial x} = -\frac{y}{r^2}.$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial u}{\partial \varphi}\frac{\partial \varphi}{\partial y} = \frac{\partial u}{\partial r}\frac{y}{r} + \frac{\partial u}{\partial \varphi}\frac{x}{r^2}.$$

Подставим:

$$x\left(\frac{\partial u}{\partial r}\frac{y}{r} + \frac{\partial u}{\partial \varphi}\frac{x}{r^2}\right) - y\left(\frac{\partial u}{\partial r}\frac{x}{r} - \frac{\partial u}{\partial \varphi}\frac{y}{r^2}\right) = 0.$$

$$\frac{\partial u}{\partial r}\left(\frac{xy}{r} - \frac{xy}{r}\right) + \frac{\partial u}{\partial \varphi}\left(\frac{x^2}{r^2} + \frac{y^2}{r^2}\right) = 0$$

$\frac{\partial u}{\partial \varphi} \cdot 1 = 0.$ ↳ $u = f(r)$ - не зависит от φ.

Ответ: $u = f(\sqrt{x^2+y^2})$.

§3 №88.2.

Преобразовать в независимые переменные: (u,v)

$x = u$
$y = uv$

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = z.$$

$u(x,y) \quad v(x,y)$

Подставим замену:

$u'_x = \frac{\partial u}{\partial x} = 1$

$u'_y = \frac{\partial u}{\partial y} = (x)'_y = 0.$

$v'_x = \frac{\partial v}{\partial x} = \left(\frac{y}{u}\right)'_x = y\left(\frac{1}{x}\right)'_x = -\frac{y}{x^2} = -\frac{w}{u^2} = -\frac{v}{u}.$

$v'_y = \frac{\partial v}{\partial y} = \left(\frac{y}{u}\right)'_y = \frac{1}{u}.$

$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial x} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}\left(-\frac{v}{u}\right).$

$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial y} = \frac{\partial z}{\partial u}\cdot 0 + \frac{\partial z}{\partial v}\left(\frac{1}{u}\right).$

Подставим:

$u\left(\frac{\partial z}{\partial u} - \frac{v}{u}\frac{\partial z}{\partial v}\right) + w\left(\frac{1}{u}\frac{\partial z}{\partial v}\right) = z.$

$u\frac{\partial z}{\partial u} - v\frac{\partial z}{\partial v} + v\frac{\partial z}{\partial v} = z$

$$u\frac{\partial z}{\partial u} = z.$$

ч.т.д. $x = f(y, z)$

Преобразовать ур-ие:

$(z - x)\frac{\partial z}{\partial x} - y\frac{\partial z}{\partial y} = 0.$

Есть:

$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$

Хотим:

$$dz = \frac{\partial z}{\partial x}\frac{\partial x}{\partial y}dy + \frac{\partial z}{\partial x}\frac{\partial x}{\partial z}dz + \frac{\partial z}{\partial y}dy$$

$$\left(1 - \frac{\partial z}{\partial x}\frac{\partial x}{\partial z}\right)dz = \left(\frac{\partial z}{\partial x}\frac{\partial x}{\partial y} + \frac{\partial z}{\partial y}\right)dy.$$

dz, dy – независимы:

$$\begin{cases} 1 = \frac{\partial z}{\partial x}\frac{\partial x}{\partial z} \\ \frac{\partial z}{\partial x}\frac{\partial x}{\partial y} = -\frac{\partial z}{\partial y} \end{cases} \quad \begin{cases} \frac{\partial z}{\partial x} = \frac{1}{\frac{\partial x}{\partial z}} \\ \frac{\frac{\partial x}{\partial y}}{\frac{\partial x}{\partial z}} = -\frac{\partial z}{\partial y}. \end{cases}$$

Подставляю:

$$+(z-x)\frac{1}{\frac{\partial x}{\partial z}} + y\frac{\frac{\partial x}{\partial y}}{\frac{\partial x}{\partial z}} = 0 \quad \Big| \ \frac{\partial x}{\partial z}$$

$$z - x + y\frac{\partial x}{\partial y} = 0. \quad \Rightarrow \quad \boxed{x = y\frac{\partial x}{\partial y} + z.}$$

§4 №51.(1).

Преобразовать к полярным координатам

$x = r\cos\varphi$

$y = r\sin\varphi$

$r = \sqrt{x^2+y^2}$ $\quad \operatorname{tg}\varphi = y/x$

$$u(x,y) = \tilde{u}(r,\varphi) = \tilde{u}(r(x,y), \varphi(x,y)).$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0. \qquad \varphi = \operatorname{arctg} \frac{y}{x}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial \varphi}\frac{\partial \varphi}{\partial x} \qquad \frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial u}{\partial \varphi}\frac{\partial \varphi}{\partial y}.$$

$$\frac{\partial r}{\partial x} = \frac{2x}{2\sqrt{x^2+y^2}} = \frac{x}{r} \qquad \frac{\partial r}{\partial y} = \frac{y}{r}.$$

$$\frac{\partial \varphi}{\partial x} = -\frac{1}{1+\frac{y^2}{x^2}} \cdot \frac{y}{x^2} = -\frac{y}{x^2+y^2} = -\frac{y}{r^2}$$

$$\frac{\partial \varphi}{\partial y} = \frac{1}{1+\frac{y^2}{x^2}} \frac{1}{x} = \frac{x}{x^2+y^2} = \frac{x}{r^2}.$$

Подставляю:

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{x}{r} - \frac{\partial u}{\partial \varphi}\frac{y}{r^2}. \qquad \frac{\partial^2 u}{\partial x^2} = \left(\frac{\partial}{\partial r}\cos\varphi - \frac{\partial}{\partial \varphi}\frac{\sin\varphi}{r}\right)\left[\frac{\partial u}{\partial r}\cos\varphi - \frac{\partial u}{\partial \varphi}\frac{\sin\varphi}{r}\right]$$

- $\frac{\partial^2 u}{\partial r^2}\cos^2\varphi$
- $-\cos\varphi\sin\varphi\,\frac{\partial}{\partial r}\frac{\partial u}{\partial \varphi}\frac{1}{r} = -\cos\varphi\sin\varphi\left(-\frac{1}{r^2}\frac{\partial u}{\partial \varphi} + \frac{1}{r}\frac{\partial^2 u}{\partial r \partial \varphi}\right)$.
- $-\frac{\sin\varphi}{r}\left(\frac{\partial}{\partial \varphi}\frac{\partial u}{\partial r}\cos\varphi\right) = -\frac{\sin\varphi}{r}\left(-\sin\varphi\frac{\partial u}{\partial r} + \cos\varphi\frac{\partial^2 u}{\partial r \partial \varphi}\right)$
- $\frac{\sin\varphi}{r}\frac{\partial}{\partial \varphi}\left(\frac{\partial u}{\partial \varphi}\frac{\sin\varphi}{r}\right) = \frac{\sin\varphi}{r^2}\frac{\partial}{\partial \varphi}\left(\frac{\partial u}{\partial \varphi}\sin\varphi\right) = \frac{\sin\varphi}{r^2}\left(\frac{\partial u}{\partial \varphi}\cos\varphi + \frac{\partial^2 u}{\partial \varphi^2}\sin\varphi\right)$

Подставляю:

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial r^2}\cos^2\varphi + \frac{\sin\varphi\cos\varphi}{r^2}\frac{\partial u}{\partial \varphi} - \frac{\sin\varphi\cos\varphi}{r}\frac{\partial^2 u}{\partial r \partial \varphi} +$$

$$+\frac{\sin^2\varphi}{r}\frac{\partial u}{\partial r}-\frac{\sin\varphi\cos\varphi}{r}\frac{\partial^2 u}{\partial r\partial\varphi}+\frac{\sin\varphi\cos\varphi}{r^2}\frac{\partial u}{\partial\varphi}+\frac{\sin^2\varphi}{r^2}\frac{\partial^2 u}{\partial\varphi^2}$$

$$\frac{\partial u}{\partial y}=\frac{\partial u}{\partial r}\frac{\partial r}{\partial y}+\frac{\partial u}{\partial\varphi}\frac{\partial\varphi}{\partial y}$$

$$\frac{\partial^2 u}{\partial y^2}=\left(\frac{\partial}{\partial r}\sin\varphi+\frac{\partial}{\partial\varphi}\frac{\cos\varphi}{r}\right)\left[\frac{\partial u}{\partial r}\sin\varphi+\frac{\partial u}{\partial\varphi}\frac{\cos\varphi}{r}\right]=$$

1. $\sin\varphi\frac{\partial}{\partial r}\left(\frac{\partial u}{\partial r}\sin\varphi\right)=\sin^2\varphi\frac{\partial^2 u}{\partial r^2}$

2. $\sin\varphi\frac{\partial}{\partial r}\left(\frac{\partial u}{\partial\varphi}\frac{\cos\varphi}{r}\right)=\sin\varphi\left(\frac{\cos\varphi}{r}\frac{\partial^2 u}{\partial r\partial\varphi}-\frac{\partial u}{\partial\varphi}\frac{\cos\varphi}{r^2}\right)$.

3. $\frac{\cos\varphi}{r}\frac{\partial}{\partial\varphi}\left[\frac{\partial u}{\partial r}\sin\varphi\right]=\frac{\cos\varphi}{r}\left(\sin\varphi\frac{\partial^2 u}{\partial r\partial\varphi}+\frac{\partial u}{\partial r}\cos\varphi\right)$

4. $\frac{\cos\varphi}{r}\frac{\partial}{\partial\varphi}\left[\frac{\partial u}{\partial\varphi}\frac{\cos\varphi}{r}\right]=\frac{\cos\varphi}{r^2}\left(\cos\varphi\frac{\partial^2 u}{\partial\varphi^2}-\frac{\partial u}{\partial\varphi}\sin\varphi\right)$.

$$\frac{\partial^2 u}{\partial y^2}=\sin^2\varphi\frac{\partial^2 u}{\partial r^2}+\frac{2\sin\varphi\cos\varphi}{r}\frac{\partial^2 u}{\partial r\partial\varphi}-\frac{2\sin\varphi\cos\varphi}{r^2}\frac{\partial u}{\partial\varphi}+$$

$$+\frac{\cos^2\varphi}{r}\frac{\partial u}{\partial r}+\frac{\cos^2\varphi}{r^2}\frac{\partial^2 u}{\partial\varphi^2}.$$

Подставляем:

$$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$$

$$\frac{\partial^2 u}{\partial r^2}\left(\sin^2\varphi+\cos^2\varphi\right)+\frac{\partial u}{\partial r}\left(\frac{\sin^2\varphi}{r}+\frac{\cos^2\varphi}{r}\right)+\frac{\partial^2 u}{\partial\varphi^2}\left(\frac{\sin^2\varphi}{r^2}+\frac{\cos^2\varphi}{r^2}\right)=0.$$

$$\boxed{\frac{\partial^2 u}{\partial r^2}+\frac{1}{r}\frac{\partial u}{\partial r}+\frac{1}{r^2}\frac{\partial^2 u}{\partial\varphi^2}=0.}$$

52.1.

$u=x-at$

$v=x+at$

$$\frac{\partial^2 z}{\partial t^2}=a^2\frac{\partial^2 z}{\partial x^2}.$$

$u(x,t);\ v(x,t)$

$$z(x,t) = \tilde{z}(u,v) = \tilde{z}(u(x,t), v(x,t)).$$

$du = dx - a\,dt$
$dv = dx + a\,dt.$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial t} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial t}.$$

$$\frac{\partial u}{\partial x} = (x - at)'_x = 1 \qquad \frac{\partial v}{\partial x} = 1$$

$$\frac{\partial u}{\partial t} = (x - at)'_t = -a \qquad \frac{\partial v}{\partial t} = a$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}$$

$$\frac{\partial z}{\partial t} = -\frac{\partial z}{\partial u}a + \frac{\partial z}{\partial v}a.$$

$$\frac{\partial^2 z}{\partial x^2} = \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v}\right)\left[\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}\right] = \frac{\partial^2 z}{\partial u^2} + 2\frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}$$

$$\frac{\partial^2 z}{\partial t^2} = \left(\frac{\partial}{\partial v}a - \frac{\partial}{\partial u}a\right)\left[\frac{\partial z}{\partial v}a - \frac{\partial z}{\partial u}a\right] = a^2\frac{\partial^2 z}{\partial v^2} + a^2\frac{\partial^2 z}{\partial u^2} - 2a^2\frac{\partial^2 z}{\partial u \partial v}$$

Подставили:

$$a^2\left(\frac{\partial^2 z}{\partial v^2} - 2\frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial u^2}\right) = a^2\left(\frac{\partial^2 z}{\partial u^2} + 2\frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}\right).$$

$$\frac{\partial^2 z}{\partial u \partial v} = 0.$$

§5 №18.3.

Исследовать на строгий экстремум

$\Delta_1 = \frac{\partial^2 f}{\partial x^2} > 0 \qquad \Delta_2 = \begin{vmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{vmatrix} > 0 \leftarrow$ min

$\Delta_1 < 0 \quad \Delta_2 > 0 \leftarrow$ max

$u = u(x;y)$

$x^3 - y^2 + u^2 - 3x + 4y + u - 8 = 0.$

$F(x,y,u) = 0$

$\frac{\partial F}{\partial x} = 3x^2 - 3 \qquad \frac{\partial u}{\partial x} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial u}} = -\frac{3x^2-3}{2u+1} = 0$

$\frac{\partial F}{\partial y} = -2y + 4 \qquad \frac{\partial u}{\partial y} = -\frac{F'_y}{F'_u} = \frac{4-2y}{2u+1} = 0$

$\frac{\partial F}{\partial u} = 2u + 1.$

$\Rightarrow$ Стационарные точки: $x^2 = 1 \quad \begin{cases} x = -1 \\ x = 1 \end{cases}$

$y = 2$

$\Rightarrow$ Точки возможные:

$(1;2) \quad (-1;2)$

$\frac{\partial^2 u}{\partial x^2} = -\frac{6x}{2u+1} \qquad \frac{\partial^2 u}{\partial y^2} = \frac{-2}{2u+1} \qquad \frac{\partial^2 u}{\partial x \partial y} = 0.$

Критерий Сильвестра:

$\Delta_2 = \begin{vmatrix} \frac{-6x}{2u+1} & 0 \\ 0 & \frac{-2}{2u+1} \end{vmatrix} = \frac{12x}{(2u+1)^2} \qquad \frac{-6x}{2u+1} = \Delta_1$

Точка строгого min, если: $\begin{cases} -\frac{6x}{2u+1} > 0 \\ \end{cases}$

Подставим точки:

$(1;2)$:

$1 - 4 + u^2 - 3 + 8 + u - 8 = 0$

$u^2 + u - 6 = 0$

$$\begin{cases} \frac{12x}{(2u+1)^2} > 0 \end{cases}$$

Строго max, если:

$$\begin{cases} \frac{-6x}{2u+1} < 0 \\ \frac{12x}{(2u+1)^2} > 0 \end{cases}$$

$D = 1 + 24 = 25$

$$\left[\begin{array}{l} -\frac{1+5}{2} = 2 \\ -\frac{1-5}{2} = -3 \end{array}\right.$$

Итого: $(1;2;2)$; $(1;2;-3)$

$(-1;2)$:

$-1 - u + u^2 + 3 + 8 + u - 8 = 0$

$u^2 + u - 2 = 0$

$$\left[\begin{array}{l} u_1 = 1 \\ u_2 = -2 \end{array}\right.$$

$(-1;2;1)$

$(-1;2;-2)$.

Подставляем:

- $(1;2;2)$:

$$\begin{cases} \frac{-6}{5} < 0 \\ \frac{12}{25} > 0 \end{cases} \leftarrow \text{max.}$$

- $(1;2;-3)$:

$$\begin{cases} \frac{-6}{-5} > 0 \\ \frac{12}{25} > 0 \end{cases} \leftarrow \text{min}$$

$(-1;2;1)$:

$\frac{6}{3} > 0$

$\frac{-12}{9} < 0$ → Нельзя определить

$(-1;2;-2)$:

$\frac{1}{-3} < 0$

$\frac{-12}{\infty} < 0$ → Нельзя.

⇒ Ответ: $(1;2;2)$ $(1;2;-3)$

21.2. Найти условие экстремума относ. условия связи:

$u = 1 - 4x - 8y$ $\quad x^2 - 8y^2 = 8.$ $\quad x^2 - 8y^2 - 8 = 0.$

Функция Лагранжа

$$L(x) = f(x) + \sum \lambda_i \varphi_i(x)$$

$$L(x) = 1 - 4x - 8y + \lambda(x^2 - 8y^2 - 8).$$

Необх. условия:

$$\begin{cases} \frac{\partial L(x^0)}{\partial x_k} \\ \varphi_i(x^0) = 0 \end{cases} \Rightarrow \begin{cases} \frac{\partial L}{\partial x} = -4 + 2\lambda x = 0 \Rightarrow \lambda = \frac{2}{x} \Rightarrow x = \frac{2}{\lambda} \\ \frac{\partial L}{\partial y} = -8 - 16\lambda y = 0 \Rightarrow \lambda = -\frac{1}{y^2} \Rightarrow y = -\frac{1}{2\lambda} \\ x^2 - 8y^2 - 8 = 0 \end{cases}$$

Подставлю: $\frac{4}{\lambda^2} - \frac{8}{4\lambda^2} - 8 = 0 \quad | \lambda^2$

$4 - 2 - 8\lambda^2 = 0 \qquad \lambda^2 = +\frac{1}{4}.$

$$\begin{cases} \lambda_1 = 1/2 \\ \lambda_2 = -1/2 \end{cases}$$

Кандидаты: $\lambda = \frac{1}{2} \quad (4; -1)$

$\lambda = -\frac{1}{2} \quad (-4; 1).$

Достаточные условия:

$d^2L = (dx\ dy)\, H \begin{pmatrix} dx \\ dy \end{pmatrix}$ H - матрица Гесса

$\frac{d^2L}{dx^2} = 2\lambda$

$\frac{d^2L}{dy^2} = -16\lambda$

$\frac{d^2L}{dx\,dy} = 0.$

$$H = \begin{pmatrix} 2\lambda & 0 \\ 0 & -16\lambda \end{pmatrix}.$$

Подставляю

$$d^2L = (2\lambda dx - 16\lambda dy)\begin{pmatrix} dx \\ dy \end{pmatrix} = 2\lambda dx^2 - 16\lambda dy^2$$

Дифференциал от ур-ия связи:

$x^2 - 8y^2 - 8 = 0$

$2x\,dx - 16y\,dy = 0$

$dx = \frac{8ydy}{x}$.

Подставим:

$d^2L = 2\lambda \frac{64}{x^2} y^2 dy^2 - 16\lambda dy^2 = 16\lambda \left(\frac{8y^2}{x^2} - 1 \right) dy^2$.

Подставим точки:

$\lambda = \frac{1}{2}(4;-1)$: $d^2L = 8\left(\frac{8}{16} - 1\right)dy^2 < 0 \Rightarrow$ усл. max.

$\lambda = -\frac{1}{2}(-4;1)$: $d^2L = -8\left(\frac{8}{16} - 1\right)dy^2 > 0 \Rightarrow$ усл. min

Подставим.

$u = 1 - 4x - 8y$

$u(4;-1) = 1 - 16 + 8 = -7$

$u(-4;1) = 1 + 16 - 8 = 9$

25.8. Найти усл. экстремум:

$u = f(x,y,z)$

$u = xy + 2xz + 2yz \quad xyz = 108. \quad \varphi(x) = xyz - 108.$

Функция Лагранжа:

$L(x) = xy + 2xz + 2yz + \lambda(xyz - 108)$

Необх. условия:

$$\begin{cases} \frac{\partial L}{\partial x} = 0 \\ \frac{\partial L}{\partial y} = 0 \\ \frac{\partial L}{\partial z} = 0 \\ xyz - 108 = 0 \end{cases} \Rightarrow \begin{cases} y + 2z + \lambda yz = 0 \quad y(1+\lambda z) = -2z \\ x + 2z + \lambda xz = 0 \quad x(1+\lambda z) = -2z \\ 2x + 2y + \lambda xy = 0 \\ xyz - 108 = 0 \end{cases}$$

2 $\frac{-2z}{1+\lambda z} + 2\frac{-2z}{1+\lambda z} + \lambda\frac{4z^2}{(1+\lambda z)^2} = 0.$

$\frac{-8}{1+\lambda z} + \lambda\frac{4z^2}{(1+\lambda z)^2} = 0 \Rightarrow 8 + 8\lambda z = 4z^2\lambda$

$z^2\lambda - 2\lambda z - 2 = 0$

$\Phi = \lambda^2 + 2\lambda$

Подставляю:

$\frac{4z^2}{(1+\lambda z)^2}$

$u = xy + 2xz + 2yz.$

Заменим x на y, y на x:

$u = yx + 2yz + 2xz. \quad yxz = 108.$

Ничего не поменялось,

Тогда рассмотрим функцию от двух переменных:

$u = x^2 + 4xz \qquad \varphi(x,z) = x^2z - 108 = 0.$

Ф-ия Лагранжа:

$L(x) = x^2 + 4xz + \lambda(x^2z - 108)$

Необходимое условия экстремума:

где $x \neq 0$

$$\begin{cases} \frac{\partial L}{\partial x} = 0 \\ \frac{\partial L}{\partial z} = 0 \\ x^2z = 108 \end{cases} \Rightarrow \begin{cases} 2x + 4z + 2x\lambda z = 0 \\ 4x + \lambda x^2 = 0 \\ x^2z = 108 \end{cases}$$

$$\begin{cases} x + 2z + \lambda xz = 0 \\ 4x + \lambda x^2 = 0 \\ x^2z = 108 \end{cases} \qquad z = \frac{108}{x^2}$$

$$\begin{cases} x + \frac{216}{x^2} + \frac{108\lambda}{x} = 0 \quad | \cdot x^2 \\ x(4 + \lambda x) = 0 \end{cases}$$

$\lambda = -\frac{4}{x}$

$x^3 + 216 + 108\lambda x = 0$

$x^3 - 432 + 216 = 0$

$x^3 = 216 \qquad x_1 = 6 \Rightarrow \lambda = -\frac{2}{3} \Rightarrow z = \frac{108}{36} = \frac{36}{12} = 3$

$\vdots 3$

Тогда: $\lambda = -\frac{2}{3}$ $(6;3)$

Найду: $d^2L = (dx\ dz)\, H \begin{pmatrix} dx \\ dz \end{pmatrix}$.

$$\frac{\partial^2 L}{\partial x^2} = 2 + 2\lambda z$$
$$\frac{\partial^2 L}{\partial z^2} = 0.$$
$$\frac{\partial^2 L}{\partial z \partial x} = 4 + 2\lambda x$$

$$d^2L = (dx\ dz) \begin{pmatrix} 2+2\lambda z & 4+2\lambda x \\ 4+2\lambda x & 0 \end{pmatrix} \begin{pmatrix} dx \\ dz \end{pmatrix} =$$

$$= (2dx + 2\lambda z dx + 4dz + 2\lambda x dz;\ 4dx + 2\lambda x dx \begin{pmatrix} dx \\ dz \end{pmatrix} =$$

$$= 2dx^2 + 2\lambda z dx^2 + 4dxdz + 2\lambda x dx dz + 4dxdz + 2\lambda x dxdz =$$

$$= (2 + 2\lambda z)dx^2 + (8 + 4\lambda x)dxdz$$

Продифференцируем ур-ие связи:

$x^2 z = 108$

$2xz\,dx + x^2 dz = 0 \qquad dz = -\frac{2xz\,dx}{x^2} = -\frac{2z\,dx}{x}.$

Подставляю:

$$d^2L = (2 + 2\lambda z)dx^2 - (8 + 4\lambda x)dx \frac{2z\,dx}{x}.$$

$$d^2L = \left(2 + 2\lambda z - \frac{16z}{x} - 8\lambda z\right)dx^2$$

$$d^2L = \left(2 - 6\lambda z - \frac{16z}{x}\right)dx^2$$

Подставляю точку: $\lambda = -\frac{2}{3}$ $(6;3)$

$$2 + 6\cdot\frac{2}{3}\cdot 3 - \frac{16\cdot 3}{6} = 2 + 12 - 8 > 0.$$

$\Rightarrow d^2L > 0 \Rightarrow$ условный min

Подставляю:

$(6;6;3)$: $u = x^2 + 4xz = 36 + 4\cdot 6\cdot 3 = 108.$

Ответ $u(6;6;3) = 108$

26.2.

Найти условные экстремумы

$u = f(x,y,z)$

$u = xyz$

$xy + yz + zx = 8$

$x + y + z = 5$.

$$\begin{cases} u = x^2 z \\ x^2 + 2xz = 8 \\ 2x + z = 5 \end{cases}$$

Функция Лагранжа:

$$L(x) = x^2 z + \lambda_1 (x^2 + 2xz - 8) + \lambda_2 (2x + z - 5)$$

$$\frac{\partial L}{\partial x} = 2xz + 2x\lambda_1 + 2z\lambda_1 + 2\lambda_2 = 0$$

$$\frac{\partial L}{\partial z} = x^2 + 2x\lambda_1 + \lambda_2 = 0$$

$$x^2 + 2xz - 8 = 0$$

$$2x + z - 5 = 0 \longrightarrow z = 5 - 2x.$$

Подставим: $x^2 + 10x - 4x^2 - 8 = 0$.

$$-3x^2 + 10x - 8 = 0$$

$$3x^2 - 10x + 8 = 0$$

$D = 25 - 24 = 1$

$x = \frac{5+1}{3} = 2. \Rightarrow z = 5 - 4 = 1$

$x = \frac{5-1}{3} = \frac{4}{3} \Rightarrow z = 5 - \frac{8}{3} = \frac{7}{3}$.

Подставим:

$(2;1)$: $4 + 4\lambda_1 + 2\lambda_1 + 2\lambda_2 = 0 \quad 2 + 3\lambda_1 + \lambda_2 = 0$

$4 + 4\lambda_1 + \lambda_2 = 0$

$2 + 3\lambda_1 = 4 + 4\lambda_1$

$\lambda_1 = -2 \Rightarrow \lambda_2 = 4.$

$(\lambda_1; \lambda_2; x; z) = (-2; 4; 2; 1)$.

Подставим:

$\left(\frac{4}{3};\frac{7}{3}\right)$: $\begin{cases} \frac{2\cdot 28}{9}+\frac{8}{3}\lambda_1+\frac{14}{3}\lambda_1+2\lambda_2=0 \\ \frac{16}{9}+\frac{8}{3}\lambda_1+\lambda_2=0 \end{cases}$ $\lambda_2=-\frac{16}{9}-\frac{8}{3}\lambda_1$

$\frac{2\cdot 28}{9}+\frac{22}{3}\lambda_1-\frac{16\cdot 2}{9}-\frac{16}{3}\lambda_1=0.$

$\frac{12\cdot 2}{9}+2\lambda_1=0.$ $\lambda_1=-\frac{12^{:3}}{9}=-\frac{4}{3}$; $\lambda_2=\frac{16}{9}$

$2dx+dz=0$

$dz=-2dx.$

$\frac{\partial^2 L}{\partial x^2}=2z+2\lambda_1$ $\frac{\partial^2 L}{\partial z^2}=0$ $\frac{\partial^2 L}{\partial x\partial z}=2x+2\lambda_1$

$$d^2L=(dx\ dz)\begin{pmatrix} 2z+2\lambda_1 & 2x+2\lambda_1 \\ 2x+2\lambda_1 & 0 \end{pmatrix}\begin{pmatrix} dx \\ dz \end{pmatrix}=$$

$$=\left((2z+2\lambda_1)dx+(2x+2\lambda_1)dz\ ;(2x+2\lambda_1)dx\right)\begin{pmatrix} dx \\ dz \end{pmatrix}=$$

$$=(2z+2\lambda_1)d^2x+(2x+2\lambda_1)dxdz+(2x+2\lambda_1)dxdz=$$

$$=(2z+2\lambda_1)d^2x+(4x+4\lambda_1)dxdz.$$

Подставляю:

$(2z+2\lambda_1)d^2x-(4x+4\lambda_1)2d^2x.=$

$=2(z+\lambda_1-4x-4\lambda_1)d^2x.=2(z-4x-3\lambda_1)d^2x.$

Подставляю точки:

$(\underset{\lambda_1}{-2};\underset{\lambda_2}{4};\underset{x}{2};\underset{z}{1})$: $1-8+6<0 \Rightarrow$ услов. max

$\left(-\frac{4}{3};\frac{16}{9};\frac{4}{3};\frac{7}{3}\right)$: $\frac{7}{3}-\frac{16}{3}+\frac{12}{3}=1 \Rightarrow$ услов min

Подставляю:

$u = x^2 z = 4$

$u = \frac{16}{9} \frac{7}{3} = \frac{112}{3}$

26.3. $u = f(x, y, z)$

$$\begin{cases} u = xy + yz \\ x^2 + y^2 = 2 \\ y + z = 2 \quad y > 0 \end{cases}$$

$x = \sqrt{2}\cos\varphi$

$y = \sqrt{2}\sin\varphi$

$z = 2 - y.$

Подставляем:

$u = \sqrt{2}\cos\varphi\,\sqrt{2}\sin\varphi + \sqrt{2}\sin\varphi\,2 - 2\sin^2\varphi =$

$u'_\varphi = 0:$ $= 2\sin\varphi\cos\varphi + 2\sqrt{2}\sin\varphi - 2\sin^2\varphi.$

$\sin 2\varphi$

$u'_\varphi = \cos 2\varphi \cdot 2 + 2\sqrt{2}\cos\varphi - 4\sin\varphi\cos\varphi = 0.$

$\cos 2\varphi = \cos^2\varphi - \sin^2\varphi = 2\cos^2\varphi - 1.$

$4\cos^2\varphi - 2 + 2\sqrt{2}\cos\varphi - 4\sin\varphi\cos\varphi = 0.$

$\sin\varphi = \frac{4\cos^2\varphi - 2 + 2\sqrt{2}\cos\varphi}{4\cos\varphi} = \cos\varphi - \frac{1}{2\cos\varphi} + \frac{\sqrt{2}}{2}$

$\sin^2\varphi = \cos^2\varphi + \frac{1}{2} + \frac{1}{4\cos^2\varphi} - 1 + \sqrt{2}\cos\varphi - \frac{\sqrt{2}}{4\cos\varphi}$

$\sin^2\varphi + \cos^2\varphi = 1.$

$2\cos^2\varphi - \frac{1}{2} + \frac{1}{4\cos\varphi} + \sqrt{2}\cos\varphi - \frac{\sqrt{2}}{4\cos\varphi} - 1 = 0. \quad | \cos^2\varphi$

$2\cos^4\varphi - \frac{3}{2}\cos^2\varphi + \frac{1}{4} + \sqrt{2}\cos^3\varphi - \frac{\sqrt{2}}{4}\cos\varphi = 0. \quad | 4$

$8\cos^4\varphi + 4\sqrt{2}\cos^3\varphi - 6\cos^2\varphi - \sqrt{2}\cos\varphi + 1 = 0.$

$\cos\varphi = \pm\frac{1}{2}$ $\qquad (x; y) = (1; 1)$

$(x, y) = (-1; 1)$

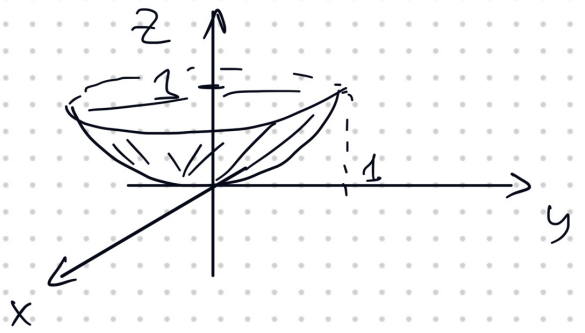
$\cos\varphi = \frac{-1 \pm \sqrt{3}}{2\sqrt{2}}$ $\qquad (x; y) = \left(\frac{-1-\sqrt{3}}{2}; \frac{-1+\sqrt{3}}{2}\right)$

31.3.

M – max; m – min значение.

$u = x + y + z$

$x^2 + y^2 \le z \le 1$.



$\frac{\partial u}{\partial x} = 1$

$\frac{\partial u}{\partial y} = 1 \Rightarrow$ Экстремумов внутри функции нет

$\frac{\partial u}{\partial z} = 1$

Подставлю z:

$u = x + y + x^2 + y^2$

$\frac{\partial u}{\partial x} = 1 + 2x = 0 \quad x = -\frac{1}{2}$

$z = \frac{1}{2}$.

$\frac{\partial u}{\partial y} = 1 + 2y = 0 \quad y = -\frac{1}{2}$.

Тогда: $u = -\frac{1}{2} - \frac{1}{2} + \frac{1}{2} = -\frac{1}{2}$.

Теперь граница: $x^2 + y^2 = 1$.

$x = \cos\varphi \quad y = \sin\varphi$.

Подставлю: $u = \cos\varphi + \sin\varphi + 1$

$u' = -\sin\varphi + \cos\varphi = 0$

$\Rightarrow \varphi = \frac{\pi}{4} + \pi k; \; k \in z$

$x = \pm\frac{\sqrt{2}}{2} \qquad u = \sqrt{2} + 1$

$y = \pm\frac{\sqrt{2}}{2} \qquad u = -\sqrt{2} + 1$

Ответ: $u\left(-\frac{1}{2}; -\frac{1}{2}; \frac{1}{2}\right) = -\frac{1}{2}$ ← min

$u\left(\frac{\sqrt{2}}{2}; \frac{\sqrt{2}}{2}; 1\right) = \sqrt{2}+1$ ← max

$u\left(-\frac{\sqrt{2}}{2}; -\frac{\sqrt{2}}{2}; 1\right) = -\sqrt{2}+1$ ← экстремум

IV. Кратные интегралы

§8 №23.

Указать функцию измеримую на м-ве, но не интегр. на нем:

$$f(x) = \begin{cases} 1, & x \in \mathbb{R} \\ 0, & x \notin \mathbb{R}. \end{cases}$$

№ 36.1.

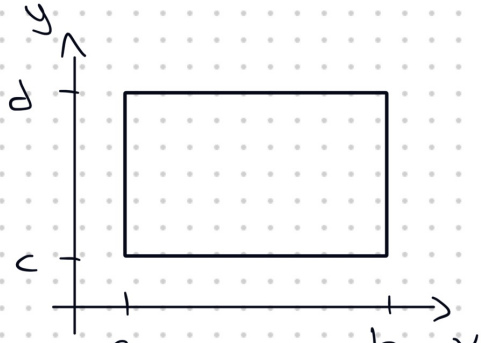
X - измерим. м-во

f - определена: $|f|$ - интегр.

f - не интегр

$$f(x) = \begin{cases} -1, & x \in \mathbb{R} \\ 1, & x \notin \mathbb{R}. \end{cases} \qquad x \in [0;1].$$

ТФ. $f(x,y)$ $P = [a,b] \times [c,d]$



Если $f(x_0 y)$ - интегр на $[c,d]$

$\forall x_0$ и $I(x) = \int_c^d f(x,y)\,dy$, то

$\exists \int_a^b \int_c^d f(x,y)\,dx\,dy$

Нет.

Пусть $[a,b] = [0,1]$; $[c,d] = [-\frac{\pi}{2}; \frac{\pi}{2}]$.

$$f(x,y) = \begin{cases} \sin y, & x \in [0,1] \cap \mathbb{Q} \\ 0, & x \in [0,1] \cap \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

$$\int_c^d f(x,y)\,dy = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} f(x,y)\,dy \begin{matrix} \xrightarrow{\mathbb{Q}} 0 \\ \xrightarrow[\mathbb{R}\setminus\mathbb{Q}]{} 0 \end{matrix}$$

$$\int_a^b 0\,dx = 0.$$

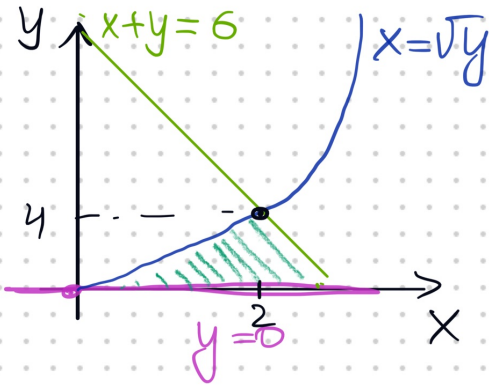
$\Rightarrow$ Повторный интеграл существует, а двойной нет.

§8 №80.3.

$y=0$; $x=\sqrt{y}$

$x+y=6$

$\iint\limits_G f(x;y)\,dx\,dy$ ← записать в виде повторных.



$$\int dx \int f(x,y)\,dy$$

$x=6-y=\sqrt{y}$

$36+y^2-12y=y$

$y^2-13y+36=0$

$D=169-4\cdot 36=25$

$\left[\begin{array}{l}\frac{13+5}{2}=9 \ (x)\\ \frac{13-5}{2}=4. \ (y)\end{array}\right.$

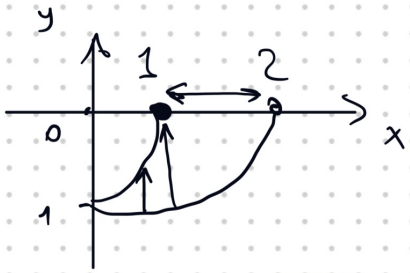
$$\int dx \int f(x,y)\,dy = \int\limits_0^2 dx \int\limits_0^{x^2} f(x,y)\,dy + \int\limits_2^6 dx \int\limits_0^{6-x} f(x,y)\,dy$$

Теперь наоборот

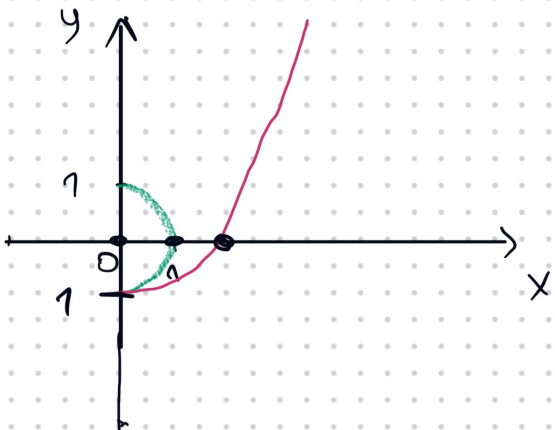
$$\int\limits_0^4 dy \int\limits_{\sqrt{y}}^{6-y} f(x,y)\,dx$$

№83.6. Изменить порядок интегрирования.

$$\int\limits_{-1}^{0} dy \int\limits_{\sqrt{1-y^2}}^{2\sqrt{y+1}} f(x,y)\,dx.$$



$\varphi(x)=\begin{cases}\sqrt{1-x^2}, & x\in[0;1]\\ 0, & x\in[1;2]\end{cases}$



$x=2\sqrt{y+1}$

$x^2=4y+4$

$y=\frac{x^2}{4}-1.$

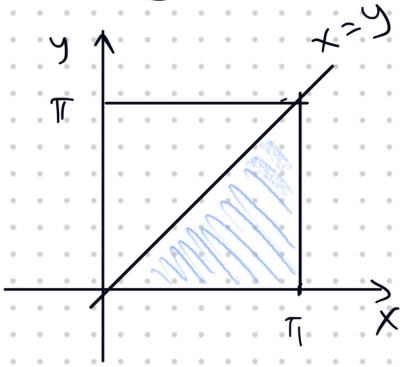
$\psi(x)$

$$\int\limits_0^2 dx \int\limits_{\frac{x^2}{4}-1}^{\varphi(x)} f(x,y)\,dy$$

№85.2.

Задание: порядок интегрирования.

$$\int_0^{\pi} dy \int_y^{\pi} \frac{\sin x}{x}\, dx = \int_0^{\pi} dx \int_0^{x} \frac{\sin x}{x}\, dy = \int_0^{\pi} \left(\frac{\sin x}{x} y \Big|_0^x \right) dx =$$



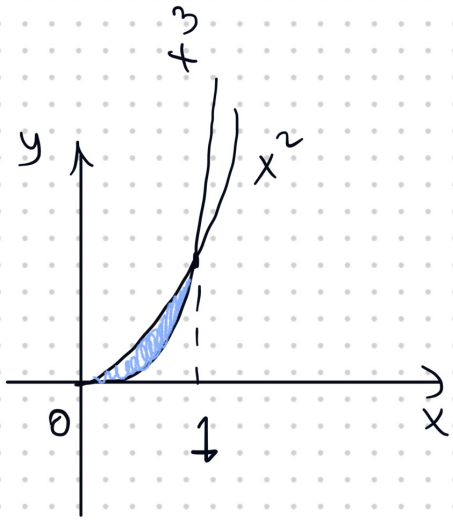
$$= \int_0^{\pi} \left(\frac{\sin x}{x} (x-0) \right) dx = -\cos x \Big|_0^{\pi} =$$

$$= 2$$

№90.2.

Вычислить двойные интегралы:

$$\iint_G \frac{y}{x^2}\, dx\, dy \quad G = \{0 < x,\ x^3 \le y \le x^2\}.$$



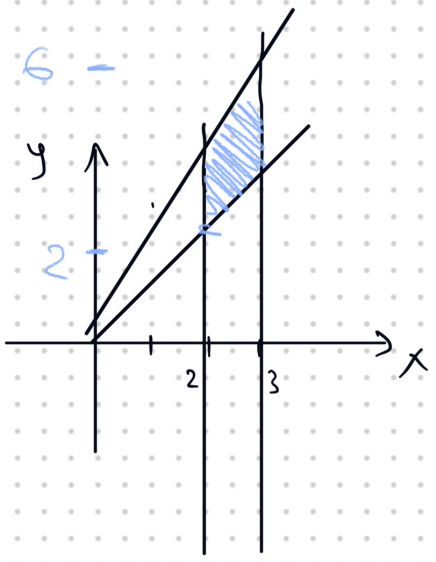
$$\int_0^1 dx \int_{x^3}^{x^2} \frac{y}{x^2} dy = \int_0^1 dx \left(\frac{y^2}{2x^2} \Big|_{x^3}^{x^2} \right) =$$

$$= \int_0^1 -\left(\frac{x^4}{2} - \frac{x^2}{2} \right) dx = -\frac{x^5}{10} + \frac{x^3}{6} \Big|_0^1 = -\frac{1}{10} + \frac{1}{6} = +\frac{2}{30} = +\frac{1}{15}.$$

№90.6.

$$\iint_G (x+2y)\, dx\, dy = \quad G\{y = x,\ y = 2x,\ x = 2,\ x = 3\}$$

$$= \int_2^3 dx \int_x^{2x} (x+2y)\, dy =$$



$$= \int_2^3 dx \left(xy + y^2 \Big|_x^{2x} \right) = \int_2^3 (2x^2 - x^2 + 4x^2 - x^2)\, dx$$

$$= \int_2^3 (4x^2)\, dx = \frac{4x^3}{3} \Big|_2^3 = \frac{4 \cdot 27}{3} - \frac{4 \cdot 8}{3} =$$

$$= \frac{19 \cdot 4}{3} = \frac{76}{3}$$

№91.6.

Вычислить двойной интеграл

$$\iint_G (2y-x)\,dx\,dy$$

$$G=\{y(y-x)\le 2,\ x(x+y)\le 3\}$$

Замена переменных

$$\int_X f(x)\,dx = \int_U f(\varphi(u))\,|J(u)|\,du$$

$$J(u)=\begin{pmatrix} \frac{\partial\varphi_1}{\partial u_1} & \cdots & \frac{\partial\varphi_1}{\partial u_n} \\ \vdots & & \vdots \\ \frac{\partial\varphi_n}{\partial u_1} & \cdots & \frac{\partial\varphi_n}{\partial u_n} \end{pmatrix}$$

Замена: $x=-u$, $y=-v$

$$J=\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}=\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \Rightarrow \det J = 1$$

$$G=\{-v(-v+u)\le 2,\ -u(-u-v)\le 3\}$$

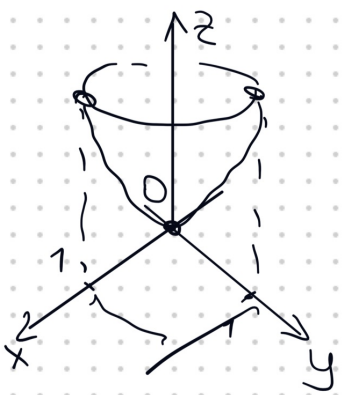
$$\iint_G (2y-x)\,dx\,dy = \iint (2(-v)+u)\cdot 1\,du\,dv = \iint (u-2v)\,du\,dv$$

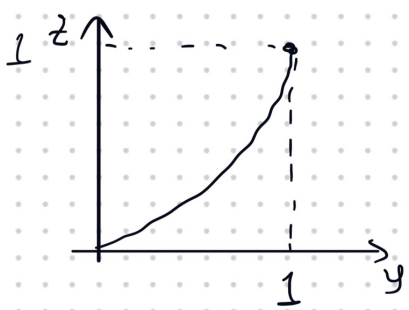
$I=-I^{*} \Rightarrow I=0.$

133.5. $\int_0^1 dx \int_0^1 dy \int_0^{x^2+y^2} f(x,y,z)\,dz.$

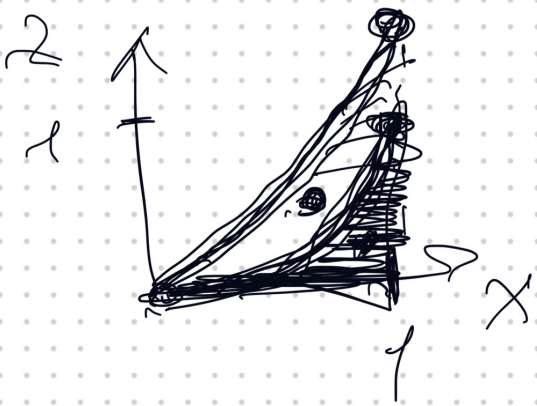
а) (x,z,y)

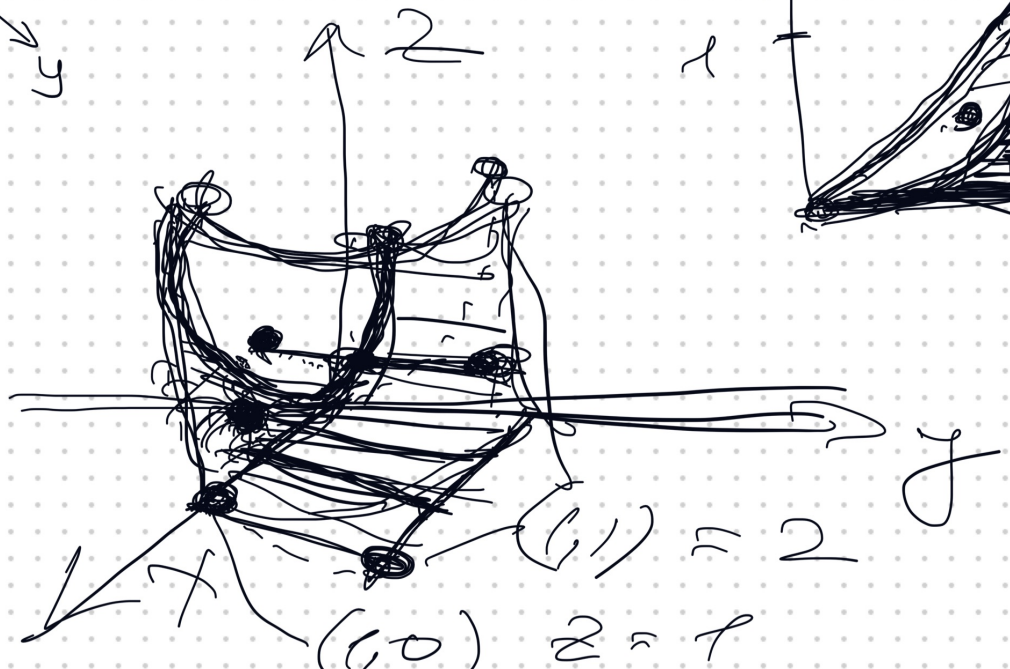
б) (z,y,x)





а) $\int dx \int dz \int f(x,y,z)\,dy$

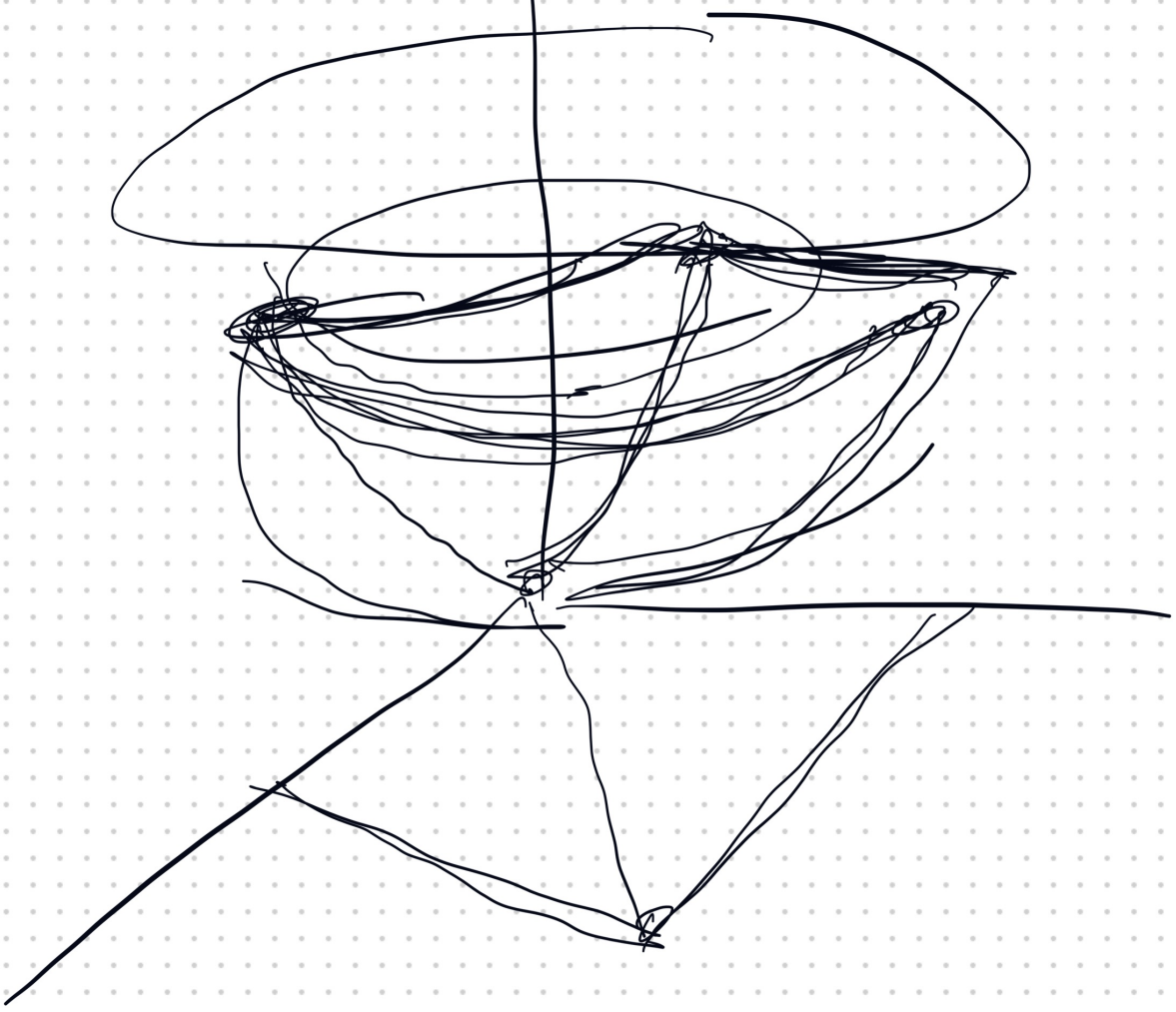




$x^2+y^2=z$

$$\int_0^1 dx \int_0^{z^2} dz \int_{\sqrt{z-x^2}}^{z^2} dy + \int_1^2 dx \int_0^{z^2} dz \int_{\sqrt{z-x^2}}^{z^2} dy$$

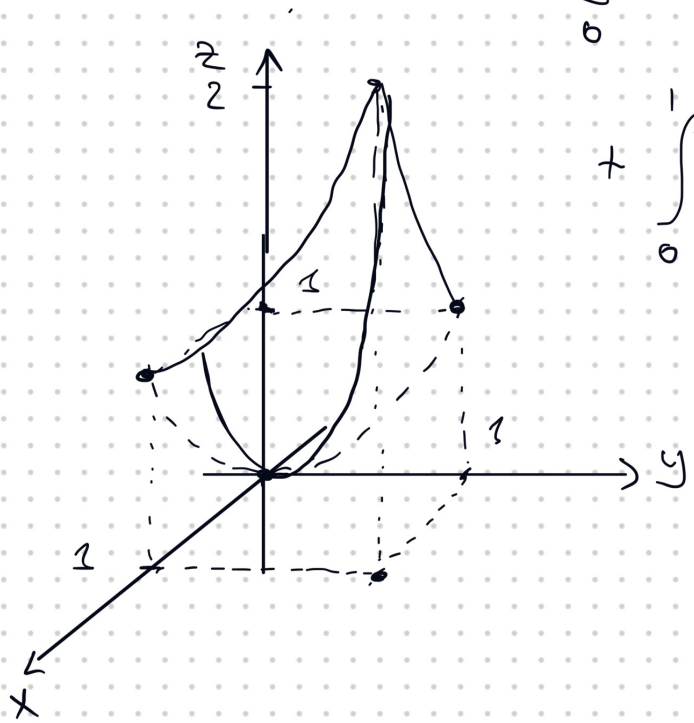
$y=\sqrt{z-x^2}$



133.5.

$$\int_0^1 dx \int_0^1 dy \int_0^{x^2+y^2} f(x,y,z)\,dz$$

1. (x,z,y).



2. (z,y,x):

$$\int_0^2 dz \int_0^{\sqrt{z}} dy \int_{\sqrt{z-y^2}}^1 f(\,)\,dx + \int$$

1. (x,z,y).

$$\int_0^1 dx \int_0^{x^2} dz \int_0^1 f(x,y,z)\,dy +$$

$$+ \int_0^1 dx \int_{x^2}^{1+x^2} dz \int_{\sqrt{z-x^2}}^1 f(\,)\,dy$$